CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Seventh Semester of B. Pharm. Examination
University Theory Examination December 2016
PH404 Pharmaceutical Analysis-III

Date: 05.12.2016, Monday **Time: 10.00 a.m to 01.00 p.m** **Maximum Marks: 80**

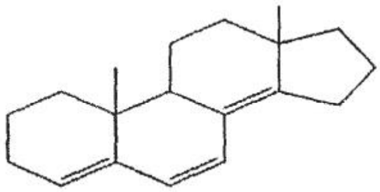
Instructions:
1. **Figures to right indicate marks.**
2. **Section wise separate answer book to be used.**
3. **Draw neat sketches wherever necessary.**

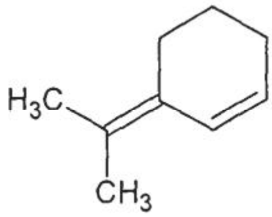
SECTION- I

Q 1 Attempt any **TWO** of the following.

A (1) Write a note on factors affecting the λ_{max} of a compound in a UV region. **05**

(2) Calculate λ_{max} of following structures. **05**





B (1) Explain the principle of FT-IR spectroscopy with suitable diagram of the instrumentation and its advantage over dispersive IR spectroscopic technique. **05**

(2) Enlist the types of UV spectrophotometers and explain double beam in Space with a suitable diagram. **05**

C (1) Write a note on factors affecting fluorescence intensity. **05**

(2) Write a note on ATR. **05**

Q 2 Attempt any **TWO** of the following.

A Write a short note on UV detectors. **10**

B (1) Explain the sample handling techniques of IR Spectroscopy. **05**

(2) Define specific absorbance, chromophore, auxochrome, phosphorescence and wavenumber. **05**

C (1) Explain principle and instrumentation of Raman spectroscopy. **05**

(2) Comment on these statements: **05**

a) Doublet is seen in IR spectra of aldehyde containing compound

b) Different types of amines show different pattern in the functional group region in the IR Spectra.

[PTO]

SECTION- II

Q 3	Attempt any **TWO** of the following.	
A	(1) Define column chromatography. Give detailed classification of column chromatography. Explain retention time and resolution.	**05**
	(2) Write a note on mobile phases and stationary phases used in HPLC.	**05**
B	(1) Draw Schematic diagram of HPLC. Write short note on pumps used in HPLC.	**05**
	(2) Discuss van Deemter equation.	**05**
C	(1) Differentiate HPTLC from TLC.	**02**
	(2) Outline basic components and steps for HPTLC technique.	**02**
	(3) Describe Silica gel GF_{254}.	**02**
	(4) How sample is applied in HPTLC technique? Explain.	**02**
	(5) Enlist applications of HPTLC.	**02**
Q 4	Attempt any **TWO** of the following.	
A	Write ideal requirements for HPLC detectors. Describe detectors commonly used in HPLC.	**10**
B	(1) What are the limitations of GC? Write about its principle and applications.	**05**
	(2) Classify GC detectors. Write about design and features of TCD.	**05**
C	(1) Give a brief note on carrier gases and columns used in GC.	**05**
	(2) Write about separation principle and applications of Ion Exchange Chromatography.	**05**